

Interpreting Remainders in Real Situations

Full groups, leftover amounts, groups needed, and estimating first

Directions

- Divide, then *stop and read the question again*. The same division can have three different answers.
- Show the division you used. Write the quotient and the remainder before you answer.
- Answer with the number the question actually asks for, and name what it counts.

1. A camp counselor has 63 water bottles to load into coolers. Each cooler holds exactly 8 bottles. How many coolers can she fill completely?

Answer: _____

2. A baker makes 95 dinner rolls and puts 6 rolls in each bag. She fills as many bags as she can. How many rolls are left over?

Answer: _____

3. A total of 138 students are going to the science museum. Each bus seats 40 students. How many buses must the school order so that every student has a seat?

Answer: _____

4. A school office has 476 pencils. Each classroom is given a set of 20 pencils.

(a) Estimate the number of classrooms by first rounding 476 to the nearest ten.

Answer: _____

(b) Now find exactly how many classrooms can be given a full set of 20 pencils.

Answer: _____

5. A librarian is shelving 1,000 books. Each shelf holds 24 books. How many shelves will be completely filled?

Answer: _____

6. A field trip takes 154 riders. Each van carries 12 riders.

- (a) How many vans are needed so that everyone gets a ride?

Answer: _____

- (b) Explain why the trip leader cannot order the number of vans that the plain quotient gives.

7. A roll of ribbon is 500 centimetres long. Each bookmark uses a piece 35 centimetres long.

- (a) How many whole bookmark pieces can be cut?

Answer: _____

- (b) How much ribbon is left over?

Answer: _____ cm

8. A food drive collected 296 cans. The volunteers pack 28 cans into each carton.

(a) Estimate the number of cartons by rounding both numbers to the nearest ten.

Answer: _____

(b) After every full carton is packed, how many cans are left over?

Answer: _____

9. A school orders 8 boxes of markers with 144 markers in each box. The markers are then shared equally among 25 classrooms.

(a) How many markers does each classroom receive?

Answer: _____

(b) How many markers are left over?

Answer: _____

10. A charity has 2,000 meal kits to ship. One pallet holds 144 kits.

(a) How many pallets will be completely full?

Answer: _____

(b) How many pallets must the charity order so that every kit ships?

Answer: _____

(c) Explain why the answers to (a) and (b) are different.
